8 **(a)** In a nuclear reaction, a uranium-235 $\left({}^{235}_{92}\text{U}\right)$ nuclide is transformed into an unstable uranium-236 nuclide $\left({}^{236}_{92}\text{U}\right)$ through bombardment by a slow-moving neutron. The unstable uranium-236 nuclide undergoes nuclear fission to form stable products of a lanthanum-139 nuclide $\left({}^{139}_{57}\text{La}\right)$ and a nuclide of bromine $\left({}^{A}_{Z}\text{Br}\right)$.

$${}^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{236}_{92}\text{U} \rightarrow {}^{139}_{57}\text{La} + {}^{A}_{Z}\text{Br} + 3\,{}^{1}_{0}\text{n} + \gamma$$

Data for the nuclei in the reaction are given in Fig. 8.1.

nucleus		mass / *u*
proton	${}^{1}_{1}\text{p}$	1.00728
neutron	${}^{1}_{0}\text{n}$	1.00866
lanthanum	${}^{139}_{57}\text{La}$	138.906
Uranium	${}^{235}_{92}\text{U}$	235.044

Fig. 8.1

(i) Determine the nucleon number *A* and proton number *Z* of the bromine nuclide.

A = ...

Z = ... [1]

(ii) Calculate the binding energy of the uranium-235 nuclide to 4 significant figures.

binding energy = ... MeV [3]

(iii) State the feature of this equation that indicates a chain reaction may be possible.

……………………………………………………………………………………………

……………………………………………………………………………………………… [1]

(b) In a laboratory source of strontium-90, the number of atoms present in the year 2013 was 2.36×10^{13}. Strontium-90 decays by emission of a β-particle and this nuclide has a half-life of 28 years.

(i) State what is a β-particle.

……………………………………………………………………………………………… [1]

(ii) Calculate the activity of the source in the year 2113.

activity = ... Bq [2]

Section B

Answer **one** question from this Section in the spaces provided.

9 **(a)** Define *simple harmonic motion*.